

work constraints while minimizing a specified objective function. In this chapter we consider the *classical* approach to economic dispatch and refer the interested reader to the citation below for details of the OPF approach.¹

We study first the most economic distribution of the output of a plant between the generators, or units, within that plant. The method that we develop also applies to economic scheduling of plant outputs for a given loading of the system without consideration of transmission losses. Next, we express transmission loss as a function of the outputs of the various plants. Then, we determine how the output of each of the plants of a system is scheduled to achieve the minimum cost of power delivered to the load.

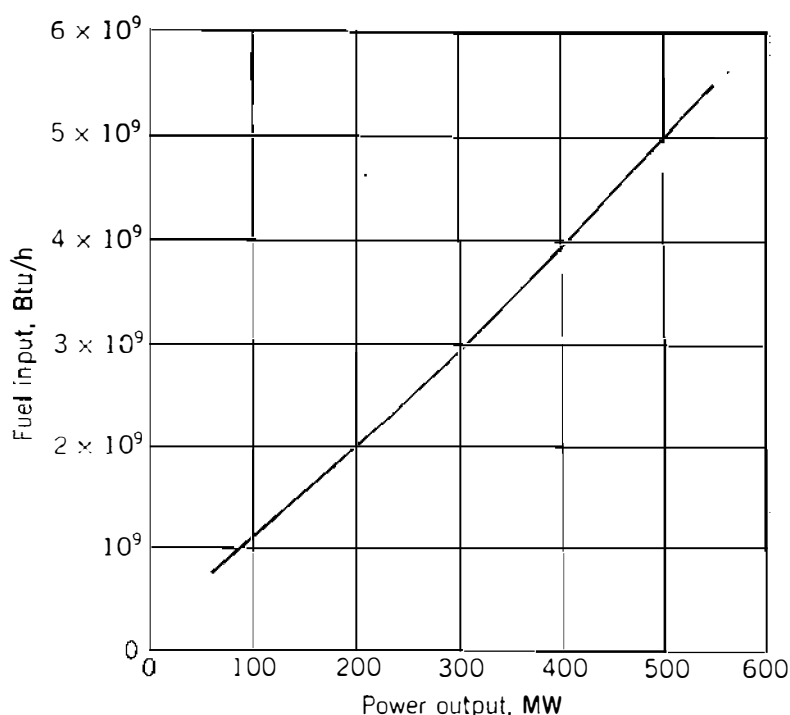
Because the total load of the power system varies throughout the day, coordinated control of the power plant outputs is necessary to ensure generation-to-load balance so that the system frequency will remain as close as possible to the nominal operating value, usually 50 or 60 Hz. Accordingly, the problem of *automatic generation control* (AGC) is developed from the steady-state viewpoint. Also, because of the daily load variation, the utility has to decide on the basis of economics which generators to start up, which to shut down, and in what order. The computational procedure for making such decisions, called *unit commitment*, is also developed at an introductory level in this chapter.

13.1 DISTRIBUTION OF LOAD BETWEEN UNITS WITHIN A PLANT

An early attempt at economic dispatch called for supplying power from only the most efficient plant at light loads. As load increased, power would be supplied by the most efficient plant until the point of maximum efficiency of that plant was reached. Then, for further increase in load the next most efficient plant would start to feed power to the system and a third plant would not be called upon until the point of maximum efficiency of the second plant was reached. Even with transmission losses neglected, this method fails to minimize cost.

To determine the economic distribution of load between the various generating units (consisting of a turbine, generator, and steam supply), the variable operating costs of the unit must be expressed in terms of the power output. Fuel cost is the principal factor in fossil-fuel plants, and cost of nuclear fuel can also be expressed as a function of output. We base our discussion on the economics of fuel cost with the realization that other costs which are a function of power output can be included in the expression for fuel cost. A typical input-output curve which is a plot of fuel input for a fossil-fuel plant in British thermal units (Btu) per hour versus power output of the unit in megawatts is shown in Fig. 13.1. The ordinates of the curve are converted to

¹H. W. Dommel and W. F. Tinney, "Optimal Power-Flow Solutions," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-87, October 1968, pp. 1866–1876.

**FIGURE 3.1**

Input-output curve for a generating unit showing fuel input versus power output.

dollars per hour by multiplying the fuel input by the cost of fuel in dollars per million Btu.

If a line is drawn through the origin to any point on the input-output curve, the slope can be expressed in millions of Btu per hour divided by the output in megawatts, or the ratio of fuel input in Btu to energy output in kilowatthours. This ratio is called the *heat rate* and its reciprocal is the *fuel efficiency*. Hence, lower heat rates imply higher fuel efficiency. Maximum fuel efficiency occurs at that point where the slope of the line from the origin to a point on the curve is a minimum, that is, at the point where the line is tangent to the curve. For the unit whose input-output curve is shown in Fig. 13.1 the maximum efficiency is at an output of approximately 280 MW, which requires an input of 2.8×10^9 Btu/h. The heat rate is 10,000 Btu/kWh, and since 1 kWh = 3412 Btu, the fuel efficiency is approximately 34%. By comparison, when the output of the unit is 100 MW, the heat rate is 11,000 Btu/kWh and the fuel efficiency is 31%.

Of course, the fuel requirement for a given output is easily converted into dollars per megawatthour. As we shall see, the criterion for distribution of the load between any two units is based on whether increasing the load on one unit as the load is decreased on the other unit by the same amount results in an increase or decrease in total cost. Thus, we are concerned with *incremental fuel cost*, which is determined by the slopes of the input-output curves of the two units. If we express the ordinates of the input-output curve in dollars per hour and let

$$f_i = \text{input to Unit } i, \text{ dollars per hour (\$/h)}$$

$$P_{gi} = \text{output of Unit } i, \text{ megawatts (MW)}$$

the incremental fuel cost of the unit in dollars per megawatthour is df_i/dP_{gi} , whereas the *average fuel cost* in the same units is f_i/P_{gi} . Hence, if the input-output curve of Unit i is quadratic, we can write

$$f_i = \frac{a_i}{2} P_{gi}^2 + b_i P_{gi} + c_i \quad \$/\text{h} \quad (13.1)$$

and the unit has incremental fuel cost denoted by λ_i , which is defined by

$$\lambda_i = \frac{df_i}{dP_{gi}} = a_i P_{gi} + b_i \quad \$/\text{MWh} \quad (13.2)$$

where a_i , b_i , and c_i are constants. The approximate incremental fuel cost at any particular output is the additional cost in dollars per hour to increase the output by 1 MW. Actually, incremental cost is determined by measuring the slope of the input-output curve and multiplying by cost per Btu in the proper units. Since mills (tenths of a cent) per kilowatthour are equal to dollars per megawatthour, and since a kilowatt is a very small amount of power in comparison with the usual output of a unit of a steam plant, incremental fuel cost may be considered as the cost of fuel in *mills per hour* to supply an additional kilowatt output.

A typical plot of incremental fuel cost versus power output is shown in Fig. 13.2. This figure is obtained by measuring the slope of the input-output curve of Fig. 13.1 for various outputs and by applying a fuel cost of \$1.30 per million Btu. However, the cost of fuel in terms of Btu is not very predictable, and the reader should not assume that cost figures throughout this chapter are applicable at any particular time. Figure 13.2 shows that incremental fuel cost is quite linear with respect to power output over an appreciable range. In analytical work the curve is usually approximated by one or two straight lines. The dashed line in the figure is a good representation of the curve. The equation of the line is

$$\lambda_i = \frac{df_i}{dP_{gi}} = 0.0126 P_{gi} + 8.9$$

so that when the power output is 300 MW, the incremental cost determined by the linear approximation is \$12.68/MWh. This value of λ_i is the approximate *additional cost* per hour of increasing the output P_{gi} by 1 MW and the *saving* in cost per hour of reducing the output P_{gi} by 1 MW. The actual incremental cost at 300 MW is \$12.50/MWh, but this power output is near the point of maximum deviation between the actual value and the linear approximation of incremental cost. For greater accuracy two straight lines may be drawn to represent this curve in its upper and lower range.

We now have the background to understand the principle of economic dispatch which guides distribution of load among the units within one or more

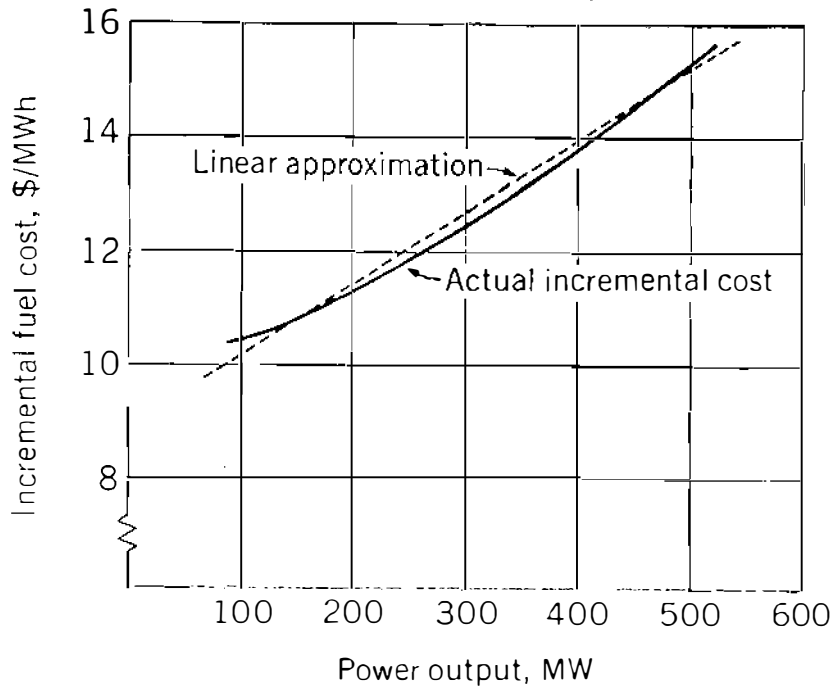


FIGURE 13.2

Incremental fuel cost versus power output for the unit whose input-output curve is shown in Fig. 13.1.

plants of the system. For instance, suppose that the total output of a particular plant is supplied by two units and that the division of load between these units is such that the incremental fuel cost of one is higher than that of the other. Now suppose that some of the load is transferred from the unit with the higher incremental cost to the unit with the lower incremental cost. Reducing the load on the unit with the higher incremental cost will result in a greater reduction of cost than the increase in cost for adding the same amount of load to the unit with the lower incremental cost. The transfer of load from one to the other can be continued with a reduction in total fuel cost until the incremental fuel costs of the two units are equal. The same reasoning can be extended to a plant with more than two units. Thus, for economical division of load between units *within* a plant, the criterion is that *all units must operate at the same incremental fuel cost*.

When the incremental fuel cost of each of the units in a plant is nearly linear with respect to power output over a range of operation under consideration, equations that represent incremental fuel costs as linear functions of power output will simplify the computations. An economic dispatch schedule for assigning loads to each unit in a plant can be prepared by:

1. Assuming various values of total plant output,
2. Calculating the corresponding incremental fuel cost λ of the plant, and
3. Substituting the value of λ for λ_i in the equation for the incremental fuel cost of each unit to calculate its output.

A curve of λ versus plant load establishes the value of λ at which each unit should operate for a given total plant load.

For a plant with two units operating under economic load distribution the λ of the plant equals λ_i of each unit, and so

$$\lambda = \frac{df_1}{dP_{g1}} = a_1 P_{g1} + b_1; \quad \lambda = \frac{df_2}{dP_{g2}} = a_2 P_{g2} + b_2 \quad (13.3)$$

Solving for P_{g1} and P_{g2} , we obtain

$$P_{g1} = \frac{\lambda - b_1}{a_1} \quad \text{and} \quad P_{g2} = \frac{\lambda - b_2}{a_2} \quad (13.4)$$

Adding these results together and then solving for λ give

$$\lambda = \left(\sum_{i=1}^2 \frac{1}{a_i} \right)^{-1} (P_{g1} + P_{g2}) + \left(\sum_{i=1}^2 \frac{1}{a_i} \right)^{-1} \left(\sum_{i=1}^2 \frac{b_i}{a_i} \right) \quad (13.5)$$

$$\text{or} \quad \lambda = a_T P_{gT} + b_T \quad (13.6)$$

$$\text{where} \quad a_T = \left(\sum_{i=1}^2 \frac{1}{a_i} \right)^{-1}$$

$$b_T = a_T \left(\sum_{i=1}^2 \frac{b_i}{a_i} \right)$$

$$\text{and} \quad P_{gT} = (P_{g1} + P_{g2})$$

is the total plant output. Equation (13.6) is a closed-form solution for λ which applies to a plant with more than two units on economic dispatch when the appropriate number of terms is added to the summations of Eq. (13.5). For instance, if the plant has K units operating on economic dispatch, then the coefficients of Eq. (13.6) are given by

$$a_T = \left(\sum_{i=1}^K \frac{1}{a_i} \right)^{-1} = \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_K} \right)^{-1} \quad (13.7)$$

$$b_T = a_T \sum_{i=1}^K \frac{b_i}{a_i} = a_T \left(\frac{b_1}{a_1} + \frac{b_2}{a_2} + \cdots + \frac{b_K}{a_K} \right) \quad (13.8)$$

and the total plant output P_{gT} is $(P_{g1} + P_{g2} + \cdots + P_{gK})$. The individual output of each of the K units is then calculated from the common value of λ

given by Eq. (13.6). If maximum and minimum loads are specified for each unit, some units will be unable to operate at the same incremental fuel cost as the other units and still remain within the limits specified for light and heavy loads. Suppose that this occurs for $K = 4$ and that the calculated value of P_{g4} violates a specified limit of Unit 4. We then discard the calculated outputs of all four units and set the operating value of P_{g4} equal to the violated limit of Unit 4. Returning to Eq. (13.6), we recalculate the coefficients a_T and b_T for the other three units and set the *effective* economic dispatch value of P_{gT} equal to the total plant load minus the limit value of P_{g4} . The resulting value of λ then governs the economic dispatch of Units 1, 2, and 3 when the actual plant output is to be increased or decreased, so long as Unit 4 remains as the only unit at a limit.

The equal-incremental-fuel-cost criterion which we have developed intuitively, and now illustrate numerically, will be established mathematically in Sec. 13.2.

Example 13.1. Incremental fuel costs in dollars per megawatthour for a plant consisting of two units are given by

$$\lambda_1 = \frac{df_1}{dP_{g1}} = 0.0080P_{g1} + 8.0 \quad \lambda_2 = \frac{df_2}{dP_{g2}} = 0.0096P_{g2} + 6.4$$

Assume that both units are operating at all times, that total load varies from 250 to 1250 MW, and that maximum and minimum loads on each unit are to be 625 and 100 MW, respectively. Find the incremental fuel cost of the *plant* and the allocation of load between units for the minimum cost of various total loads.

Solution. At light loads Unit 1 will have the higher incremental fuel cost and operate at its lower limit of 100 MW for which df_1/dP_{g1} is \$8.8/MWh. When the output of Unit 2 is also 100 MW, df_2/dP_{g2} is \$7.36/MWh. Therefore, as plant output increases, the additional load should come from Unit 2 until df_2/dP_{g2} equals \$8.8/MWh. Until that point is reached, the incremental fuel cost λ of the *plant* is determined by Unit 2 alone. When the plant load is 250 MW, Unit 2 will supply 150 MW with df_2/dP_{g2} equal to \$7.84/MWh. When df_2/dP_{g2} equals \$8.8/MWh,

$$0.0096P_{g2} + 6.4 = 8.8$$

$$P_{g2} = \frac{2.4}{0.0096} = 250 \text{ MW}$$

and the total plant output P_{gT} is 350 MW. From this point on the required output of each unit for economic load distribution is found by assuming various values of P_{gT} , calculating the corresponding plant λ from Eq. (13.6), and substituting the value of λ in Eqs. (13.4) to compute each unit's output. Results are shown in Table 13.1. When P_{gT} is in the range from 350 to 1175 MW, the plant λ is determined by Eq. (13.6). At $\lambda = 12.4$ Unit 2 is operating at its upper limit and additional load

TABLE 13.1
The plant λ and outputs of each unit
for various values of total output P_{gT} for Example 13.1

Plant		Unit 1	Unit 2
P_{gT} , MW	λ , \$/MWh	P_{g1} , MW	P_{g2} , MW
250	7.84	100 [†]	150
350	8.80	100 [†]	250
500	9.45	182	318
700	10.33	291	409
900	11.20	400	500
1100	12.07	509	591
1175	12.40	550	625 [†]
1250	13.00	625	625 [†]

[†]Indicates the output of the unit at its minimum (or maximum) limit and the plant λ is then equal to the incremental fuel cost of the unit not at a limit.

must come from Unit 1, which then determines the plant λ . Figure 13.3 shows plant λ plotted versus plant output.

If we wish to know the distribution of load between the units for a plant output of 500 MW, we could plot the output of each individual unit versus plant output, as shown in Fig. 13.4, from which each unit's output can be read for any plant output. The correct output of each of many units can be easily computed from Eq. (13.6) by requiring all unit incremental costs to be equal for any total plant output. For the two units of the example, given a total output of 500 MW,

$$P_{gT} = P_{g1} + P_{g2} = 500 \text{ MW}$$

$$a_T = \left(\frac{1}{a_1} + \frac{1}{a_2} \right)^{-1} = \left(\frac{1}{0.008} + \frac{1}{0.0096} \right)^{-1} = 4.363636 \times 10^{-3}$$

$$b_T = a_T \left(\frac{b_1}{a_1} + \frac{b_2}{a_2} \right) = a_T \left(\frac{8.0}{0.008} + \frac{6.4}{0.0096} \right) = 7.272727$$

and then for each unit

$$\lambda = a_T P_{gT} + b_T = 9.454545 \text{ $/MWh}$$

which yields

$$P_{g1} = \frac{\lambda - b_1}{a_1} = \frac{9.454545 - 8.0}{0.008} = 181.8182 \text{ MW}$$

$$P_{g2} = \frac{\lambda - b_2}{a_2} = \frac{9.454545 - 6.4}{0.0096} = 318.1818 \text{ MW}$$

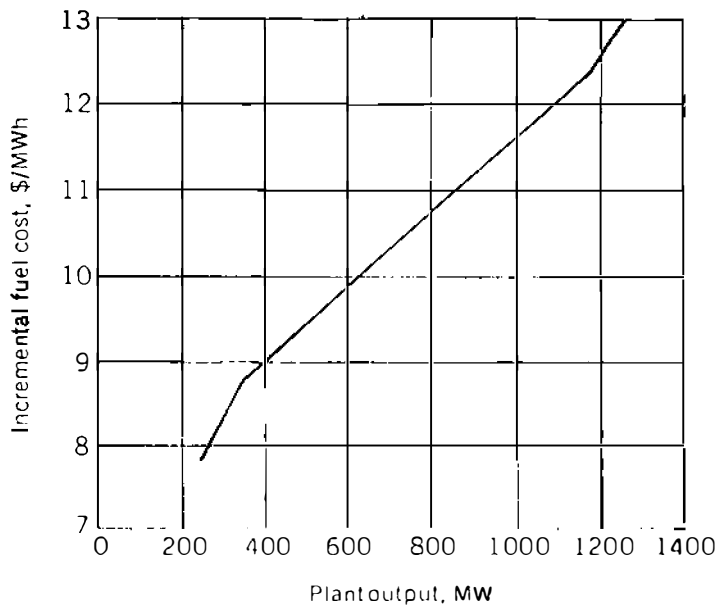


FIGURE 13.3

Incremental fuel cost versus plant output with total plant load economically distributed between units, as found in Example 13.1.

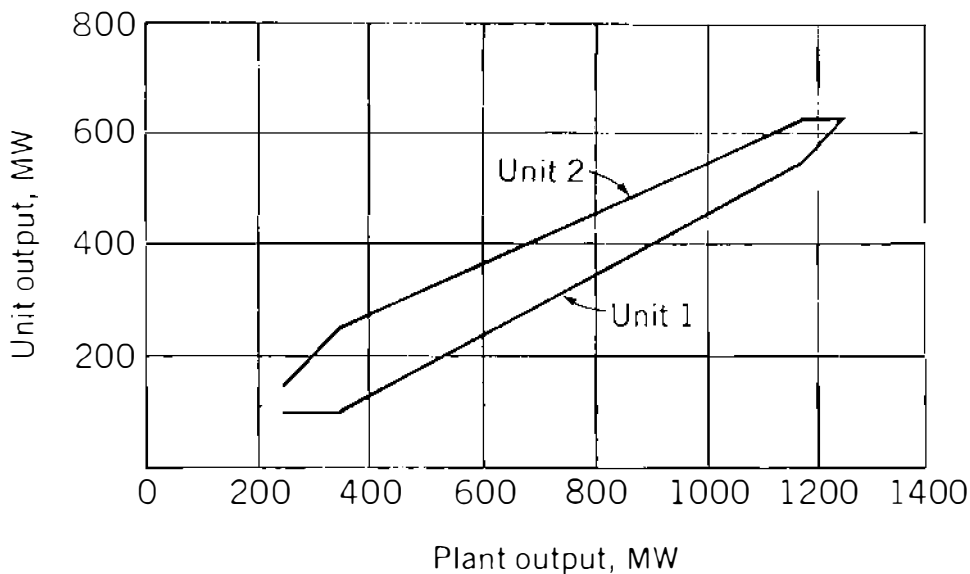


FIGURE 13.4

Output of each unit versus plant output for economical operation of the plant of Example 13.1.

Such accuracy, however, is not necessary because of the uncertainty in determining exact costs and the use of an approximate equation in this example to express the incremental costs.

The savings effected by economic distribution of load rather than some arbitrary distribution can be found by integrating the expression for incremental fuel cost and by comparing increases and decreases of cost for the units as load is shifted from the most economical allocation.

Example 13.2. Determine the saving in fuel cost in dollars per hour for the economic distribution of a total load of 900 MW between the two units of the plant described in Example 13.1 compared with equal distribution of the same total load.

Solution. Table 13.1 shows that Unit 1 should supply 400 MW and Unit 2 should supply 500 MW. If each unit supplies 450 MW, the increase in cost for Unit 1 is

$$\int_{400}^{450} (0.008P_{g1} + 8) dP_{g1} = \left(0.004P_{g1}^2 + 8P_{g1} + c_1 \right) \Big|_{400}^{450} = \$570 \text{ per hour}$$

The constant c_1 cancels when we evaluate at the two limits. Similarly, for Unit 2

$$\int_{500}^{450} (0.0096P_{g2} + 6.4) dP_{g2} = \left(0.0048P_{g2}^2 + 6.4P_{g2} + c_2 \right) \Big|_{500}^{450} = -\$548 \text{ per hour}$$

The negative sign indicates a decrease in cost, as we expect for a decrease in output. The net increase in cost is $\$570 - \$548 = \$22$ per hour. The saving seems small, but this amount saved every hour for a year of continuous operation would reduce fuel cost by \$192,720 for the year.

The saving effected by economic distribution of load justifies devices for controlling the loading of each unit automatically. We shall consider automatic control of generation later in this chapter. But first let us investigate the problem of coordinating transmission losses in the economic distribution of load between plants.

13.2 DISTRIBUTION OF LOAD BETWEEN PLANTS

In determining the economic distribution of load between plants, we encounter the need to consider losses in the transmission lines. Although the incremental fuel cost at one plant bus may be lower than that of another plant for a given distribution of load between the plants, the plant with the lower incremental cost at its bus may be much farther from the load center. The losses in transmission from the plant having the lower incremental cost may be so great that economy may dictate lowering the load at the plant with the lower incremental cost and increasing it at the plant with the higher incremental cost. Thus, we need to coordinate transmission loss into the scheduling of the output of each plant for maximum economy at a given level of system load.

For a system with K generating units let

$$f = f_1 + f_2 + \cdots + f_K = \sum_{i=1}^K f_i \quad (13.9)$$

where f is the *cost function* giving the total cost of all the fuel for the entire system and is the sum of the fuel costs of the individual units $f_1, f_2, \dots, f_K$. The